Computer Vision and Autonomous Vehicles

W7D2 Haystack Retrieval Practice Document

Computer vision

Computer vision enables computers to interpret images and video. Common tasks include image classification, object detection, image segmentation, and tracking objects across video frames.

Object detection

Object detection identifies objects and estimates their locations, often using bounding boxes and class labels. Performance depends on the quality and variety of the training data and on the conditions during use.

Autonomous navigation

An autonomous vehicle uses sensors, perception software, localization, mapping, planning, and control to move through its environment. It must estimate where it is and choose a safe path.

Safety

Vehicles may combine cameras, radar, lidar, GPS, and inertial sensors. Testing must consider poor lighting, weather, occlusion, unusual road layouts, and unexpected actions by other road users.

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